

UChicago REU Notes Week 4

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Remark 0.1. Assume everything is compactly generated, so we have a natural homeomorphism $\mathcal{U}(X \times Y, Z) \cong \mathcal{U}(X, \mathcal{T}(Y, Z))$, and $\mathcal{T}(X \wedge Y, Z) \cong \mathcal{T}(X, \mathcal{T}(Y, Z))$. Here, $\mathcal{U}$ are compactly generated spaces, $\mathcal{T}$ are based spaces.

For basic reviews of G -spaces, I'll type up the notes later.

Here, we also assume V is a finite dimensional $\mathbb{R}$ -inner product space with a G -representation.

1 Basics of Equivariant Homotopy

2 Motivation of Categorical Aspect

Goal: Understand to construct genuine G -spectra. A naive G -spectra is a spectra with a G -action (suposedly the structure maps are all G -equivariant + homotopy equivalence, or homeomorphism). Seems like we require extra structures on the spectra. One of the distinction seems to be: naive G -spectra represents $\mathbb{Z}$ -graded cohomology theories, genuine G -spectra are used to represents cohomology theories for real representations (so it's also encoded using representations). I particular, understanding what is the equivariant infinite loop space machine (construct from G -space / category, forming a G -spectra).

Definition 2.1. Given a category $\mathcal{V}$, $X \in \mathcal{V}$ is a G -object if it has a group homomorphism $G \rightarrow \text{Aut}(X)$.

$\mathcal{V}_G$ denotes the category of G -objects, with morphisms being in $\mathcal{V}$. Each $\mathcal{V}(X, Y)$ has a natural G -action by conjugation, where $g \in G$ acts as $f \mapsto g \circ f \circ g^{-1}$.

$G\mathcal{V} \subseteq \mathcal{V}_G$ denotes the subcategory where homsets are limited to G -maps (so $G\mathcal{V}(X, Y) := \mathcal{V}(X, Y)^G$, fixed points of the homsets). Another notation is $G\mathcal{V} := (\mathcal{V}_G)^G$.

Definition 2.2. Given $X \in G\mathcal{T}$ (based G -space), assume inclusion of basepoint $* \rightarrow X$ is nondegenerated in the G -case, meaning it's a G -cofibration (satisfying G -homotopy extension property for G -equivariant

maps, abbreviated as G -maps). In particular, all the homotopies in the extension property is required to be a G -map at all time t (another way of phrasing, interval I has trivial action, so action at slice $X \times t$ is equivalent to action on $X \times I$).

By chasing through the homotopy extension diagram (by restricting all homotopies to be G -homotopy), then inclusion of basepoint $*$ $\rightarrow X^H$ becomes a cofibration with respect to G -maps (Note: Not necessarily a G -cofibration, because in general X^H is not closed under G -action). However, since all G -space is naturally an H -space (so is homotopy), then we can understand this as “ H -cofibration”, as H acts trivially on X^H :0

Definition 2.3. A G -map $f : X \rightarrow Y$ is a weak equivalence of G -spaces (or G -equivalence), if the fixed point maps $f^H : X^H \rightarrow Y^H$ are weak homotopy equivalences for all subgroups $H \leq G$.

Similar G -whiteheads theorem states if X, Y are G -CW complexes, then f is also a G -homotopy equivalence.

Definition 2.4. A small category $\mathcal{C}$ is a *topological category*, if the set of objects Ob and set of all morphisms Mor are topological spaces, such that the following four maps are continuous:

- Identity map $i : \text{Ob} \rightarrow \text{Mor}$ by $i(x) := \text{id}_x$ is a continuous map.
- Source map $s : \text{Mor} \rightarrow \text{Ob}$ by $s(f : X \rightarrow Y) := X$ is continuous.
- Target map $t : \text{Mor} \rightarrow \text{Ob}$ by $t(f : X \rightarrow Y) := Y$ is continuous.
- Composition $\circ : \text{Mor} \times_{t,s} \text{Mor} \rightarrow \text{Mor}$ is continuous.

Which, the last one is collecting all pair (X, Y) with morphism $f : X \rightarrow Y$ between them.

In particular, s, t are retracts of i , showing i is an embedding. (check <https://mathoverflow.net/questions/85717/segals-original-definition-of-a-topological-category>)

There is another notion enriched over $\mathcal{T}$.

Notation 2.5. Let Cat denotes the category of small topological categories, which Cat_G denotes the topological G -categories (meaning G acts on both Ob, Mor such that $i, s, t, \circ$ are all G -maps), an equivalent formulation is a group homomorphism $G \rightarrow \text{Aut}_{\text{Cat}}(\mathcal{C}, \mathcal{C})$, by checking the definition.

Part of the reason we endow this definition, is so that for subgroup $H \leq G$, the H -fixed category $\mathcal{C}^H$ will make sense, namely we need the action on Ob, Mor to be compatible.

Remark 2.6. Given the classifying space functor B , it realizes any topological category $\mathcal{C}$ into BC .

Question: What are the comparison between the two spaces?

Which, for some reason it's product preserving, and it sends objects in Cat to $\mathcal{U}$, $G\text{Cat}$ to $G\mathcal{U}$. In particular, it preserves the operad action (as preserving product also preserves the action).

3 E_V Operads, V -fold Loop G -Spaces

3.1 Little Disks / Steiner Operad

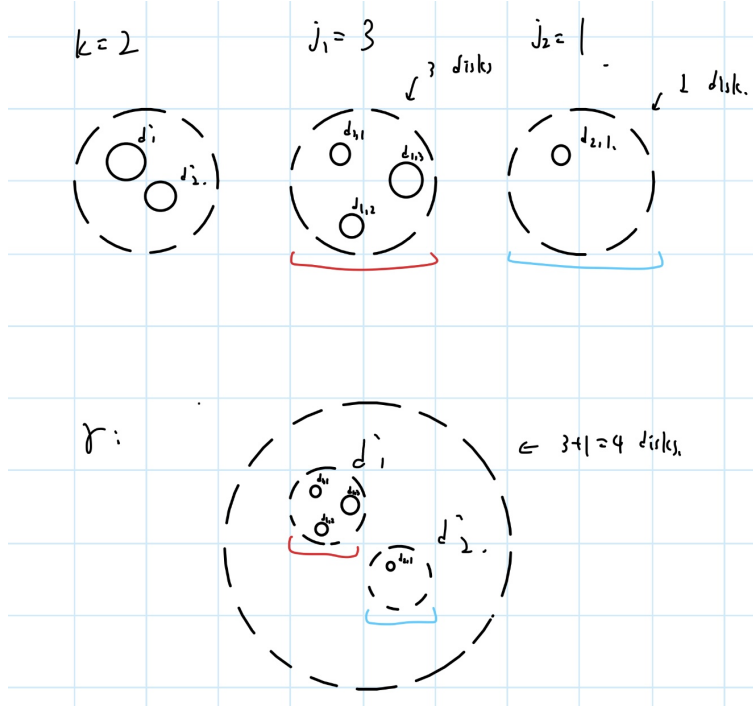
Definition 3.1. Given $D(V) \subset V$ the open unit disk (all $v \in V$ with $\|v\| < 1$), a *little V -disk* is an affine map $d : D(V) \rightarrow D(V)$, meaning all $d(v) = rv + v_0$, where $r \in [0, 1)$ and $v_0 \in D(V)$ represents the radius and the center.

Define group action $g \in G$ on collection $(gd)(v) := rv + gv_0$, which forms an evident G -action with restriction of action $G \rightarrow O(V)$ (the real unitary group of V).

Define the *little disk operad* by defining $\mathcal{D}_V(j) := \{(d_1, \dots, d_j) \mid d_i \text{ shares empty image intersection}\}$. Geometrically, it's all the possible arrangement of j non-overlapping disks in $D(V)$. Define the evident structure map $\gamma : \mathcal{D}_V(k) \times \mathcal{D}_V(j_1) \times \dots \times \mathcal{D}_V(j_k) \rightarrow \mathcal{D}_V(j)$ as follow:

$$\gamma((d'_1, \dots, d'_k); (d_{1,1}, \dots, d_{1,j_1}), \dots, (d_{j_k,1}, \dots, d_{j_k,j_k})) = ((d'_1 \circ d_{1,1}, \dots, d'_1 \circ d_{1,j_1}); \dots; (d'_k \circ d_{j_k,1}, \dots, d'_k \circ d_{j_k,j_k}))$$

Geometrically, it's as follow (for an example):



This is obvious a Σ -operad (as we defined it to be function composition). There is a correspondance of this definition to configuration space: Let $F(V, j) \subset V^j$ be the ordered configuration space of j -tuples of distinct points of V , with G acts on V^j diagonally (which restricts to an action on $F(V, j)$).